

Problem 16.12

Find the present value of payments of \$200 every six months starting immediately and continuing through four years from the present, and \$100 every six months thereafter through ten years from the present, if $i^{(2)} = 0.06$.

Solution

Since $i^{(2)} = 0.06$, the effective six-month interest rate is

$$j = \frac{0.06}{2} = 0.03.$$

The \$200 payments are made every six months starting immediately for the first four years. This gives

$$4(2) = 8$$

payments as an annuity due, so their present value is

$$200\ddot{a}_{\overline{8}|}.$$

The \$100 payments are made every six months thereafter through ten years from the present. This gives

$$(10 - 4)(2) = 12$$

payments, deferred for 8 six-month periods. Their present value is

$$100\ddot{a}_{\overline{12}|}(1.03)^{-8}.$$

Therefore, the total present value is

$$PV = 200\ddot{a}_{\overline{8}|} + 100\ddot{a}_{\overline{12}|}(1.03)^{-8}.$$

Using

$$\ddot{a}_{\overline{n}|} = (1 + j) \left(\frac{1 - (1 + j)^{-n}}{j} \right),$$

we get

$$\begin{aligned} PV &= 200(1.03) \left(\frac{1 - (1.03)^{-8}}{0.03} \right) \\ &\quad + 100(1.03) \left(\frac{1 - (1.03)^{-12}}{0.03} \right) (1.03)^{-8}. \end{aligned}$$

Thus,

$$\begin{aligned} PV &\approx 1446.0566 + 809.3516 \\ &\approx 2255.4082. \end{aligned}$$

Hence, the present value is

$$\boxed{\$2,255.41}.$$